Market and Patent Research Report.

ProdDev SS25 - Group A

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1 Introduction

Invasive alien plant species such as Heracleum mantegazzianum, Lupinus polyphyllus, and Jacobaea vulgaris pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing. Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2 Market Overview and Technological Landscape

2.1 RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum.*
- **Advantages:**
  - Non-chemical and environmentally safe.
  - Minimal soil disturbance.
  - Suitable for robotic integration. Carbon Robotics
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.

Figure 1: Source: RootWave

Video Demonstration:

- RootWave Demonstration 1
- RootWave Demonstration 2

2.2 Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO_2 lasers for precision weed eradication.
- **Target Species:** More suitable for row crops. Not designed specifically for tall or toxic plants like hogweed.

Figure 2: Source: RootWave

- **Video Demonstration:** LaserWeeder Overview

2.3 Odd. Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Target Species:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.

Figure 3: Source: RootWave

2.4 Drone and Aerial Detection Platforms (DK, DE, UK)

- **Developer**: University of Connecticut (BirdHabitatBot)
- **Technology:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Advantages:** Ideal for pre-mapping and targeting robotic ground intervention.

Figure 4: Source: RootWave

2.5 FarmWise – Titan FT-35 (USA)

- **Manufacturer:** FarmWise
- **Technology:** A self-driving robotic cultivator with vision and mechanical blade system for precision weeding. Optional fertilizer or pesticide application.
- **Target Species:** Broadleaf weeds in vegetable crops.
- **Use Cases:** Commercial vegetable farms.

2.6 Naïo Technologies – Dino (France)

- **Manufacturer:** Naïo Technologies
- **Technology:** GPS navigation, laser scanners, vision-based selective mechanical weeding.
- **Target Species:** Weeds between and near crop rows.
- **Use Cases:** Organic farms requiring non-chemical weeding.

2.7 ecoRobotix – AVO (Switzerland)

- **Manufacturer:** ecoRobotix
- **Technology:** Solar-powered robot with precision herbicide spray using machine learning.
- **Target Species:** Isolated weeds among broadleaf crops.
- **Use Cases:** Sustainable farming.

2.8 FarmDroid – FD20 (Denmark)

- **Manufacturer:** FarmDroid
- **Technology:** GPS-based seeding and blind mechanical weeding.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** Organic grain and vegetable farms.

2.9 Small Robot Company – "Tom & Dick" (UK)

- **Manufacturer:** Small Robot Company
- **Technology:** Tom scouts and maps, Dick applies electric shock to kill weeds.
- **Target Species:** Broadleaf weeds in high-value crops.
- **Use Cases:** Precision agriculture.

2.10 AutoWeed – Lantana Targeting Robot (Australia)

- **Developer:** James Cook University
- **Technology:** AI and cameras detect and spray herbicide on *Lantana camara*.
- **Target Species:** *Lantana camara*.
- **Use Cases:** Biodiversity protection.

2.11 BirdHabitatBot – Forest Invasives Robot (USA)

- **Developer:** University of Connecticut
- **Technology:** Drone-GPS mapping + ground robot to remove invasive shrubs.
- **Target Species:** *Japanese barberry*, *multiflora rose*.
- **Use Cases:** Forest conservation.

2.12 Bosch Deepfield – BoniRob (Germany)

- **Developer:** Bosch Research
- **Technology:** Vision-guided pneumatic "stamping" system.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** High-speed non-chemical cultivation.

2.13 Hydrilla Hunter – Robotic Boat (USA)

- **Developer:** Northeastern University & CAES
- **Technology:** Hyperspectral aquatic detection and robotic navigation.
- **Target Species:** *Hydrilla*, *Eurasian watermilfoil*.
- **Use Cases:** Freshwater habitat protection.

3 Patent Research Overview

3.1 EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in *Heracleum mantegazzianum.*
- **Significance:** Basis for robotic puncturing adaptation.
- **Link:** EP3804483A1

3.2 WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon Robotics
- **Description:** Robotic platform guiding laser source for weed removal.
- **Limitation:** Not optimized for toxic plants or non-row fields.
- **Link:** WO2021055485A1

3.3 US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using sensors and mechanical tools.
- **Potential Use:** Adaptable for invasive species.
- **Link:** US20170238460A1

3.4 EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH
- **Description:** Electrode-based high-voltage weed inactivation.
- **Relevance:** Integrable into autonomous platforms.
- **Link:** EP3744173A1

3.5 DE102015209879A1 – Device for Damaging Weeds

- **Assignee:** Bosch GmbH
- **Description:** Tool with classification and positioning unit for targeted weed damage.
- **Significance:** Precise, rapid chemical-free weed control.
- **Link:** DE102015209879A1

3.6 DE4039797A1 – Weed Control Device

- **Assignee:** Robert Bosch GmbH
- **Description:** Continuous weed destruction system interrupted only when crop detected.
- **Significance:** Efficient for automated systems.
- **Link:** DE4039797A1